

Bill of materials IDK+(TDK) 'Retrophone' Rev1 (july 2023) & Rev2 (march 2025) - Arnoud Whizzbizz.com

Designator	Package	Quantity	Designation	Comment, Link
R12,R5	Axial DIN0207 L6.3mm P10.16mm	2	1K	
R3	Axial DIN0207 L6.3mm P10.16mm	1	330	
R17,R16,R8	Axial DIN0207 L6.3mm P10.16mm	3	2K2	
D3,D8,D7,D6	LED_D3.0mm	4	LED 3mm	
U5	Arduino_Pro_Mini	1	Pro Mini 328	35 pin Arduino Pro Mini module with FTDI programming port
SW1	SW_Push_2P2T_7x7	1	POWER	7x7mm DPDT with lock - Take care of mounting orientation!
U1	DFPlayer_Mini	1	DFPlayerMini	DFPlayer Mini module. Use MP3-TF-16P with the blue LED!
CP2,CP1	CP_Radial_D5.0mm_P2.50mm	2	10uF	
C3,C5	C_Disc_D3.0mm_W1.6mm_P2.50mm	4	100nf	
F1	Fuse_8.5x4	1	500mA	
U3	TO-220-3_Horizontal_TabUp_Heatsink	1	LM7805_TO220	LM7805 with alu heatsink 22x15 mm, 10mm high for TO-220 package
Q1	TO-92_Inline	1	BC547	
C2	CP_Radial_D5.0mm_P2.50mm	1	220uF	
U2	DIP-14_W7.62mm_LongPads	1	74LS14	DIP14 TTL Hex Schmitt Trigger Inverters
D2,D4	D_DO-41_SOD81_P7.62mm_Horizontal	2	1N5817	Schottky diodes
C7	CP_Radial_D5.0mm_P2.50mm	1	470uF	
C8	CP_Radial_D5.0mm_P2.50mm	1	3u3F	
C4	CP_Radial_D5.0mm_P2.50mm	1	100uF	
R10	Axial DIN0207 L6.3mm P10.16mm	1	10K	
R14,R15	Axial DIN0207 L6.3mm P10.16mm	2	10K	
R2	Axial DIN0207 L6.3mm P10.16mm	1	27K	
R4	Axial DIN0207 L6.3mm P10.16mm	1	2K2	
R1	Axial DIN0207 L6.3mm P10.16mm	1	330	
D5	D_DO-35_SOD27_P7.62mm_Horizontal	1	1N4148	Rotated cathode designator on silk screen of pcb Rev1! Rotate 180 degree
J1	BarrelJack_Horizontal	1	Barrel_Jack_Switch	DC input for 9v adaptor (5.5mm with 2.5mm pin)
J3	PinHeader_1x02_P2.54mm_Vertical	1	Phone	2-pin header for connecting a/b of the phone connector
D1	D_DO-35_SOD27_P7.62mm_Horizontal	1	1N4001	Rotated cathode designator on silk screen of pcb Rev1! Rotate 180 degree
J2	USB_B_OST_USB-B1HSxx_Horizontal	1	USB_B	USB-B type pcb connector (optional)
R7	Wire bridge (Resistor has been removed)	1	Wire	Only on pcb Rev1 (july 2023), was corrected in pcb Rev2 (march 2025)

Optional momentary-action switch SW2 Util/Config button with LED D6

Designator	Package	Quantity	Designation	Comment, Link
SW2	SW_Push_2P2T_7x7	1	UTIL/CONFIG	7x7mm DPDT without lock - Take care of mounting orientation!
R13	Axial DIN0207 L6.3mm P10.16mm	4	2K2	
D6	LED_D3.0mm	4	LED 3mm	Util/Config LED

Optional DTMF ('TDK') decoding (tone dialing and modern phones)

Designator	Package	Quantity	Designation	Comment, Link
Y1	Crystal_HC18-U_Vertical	1	3.579545 MHz	May be classic high model, or low passive resonator model
U4	DIP-18_W7.62mm	1	MT8870	DIP18 version of the MT8870 or MV8870 DTMF decoder
C1,C6	C_Disc_D3.0mm_W1.6mm_P2.50mm	4	100nf	
R6,R9	Axial DIN0207 L6.3mm P10.16mm	2	100K	
R11	Axial DIN0207 L6.3mm P10.16mm	1	330K	

Casing, 3D-prints, additional assembly materials

Designator	Package	Quantity	Designation	Comment, Link
PCB	Printed Circuit Board	1	Rev1 or Rev2	Printed Circuit Board (Gerber files on GitHub)
3DCASE	3D printed case (either for RJ11 or Dutch PTT connector)	1		STL files of (wall-)case and lid on GitHub
HTSNK	Heatsink for TO-220	1		Aluminum heatsink 22x15 mm, 10mm high for TO-220
SPACER	Aluminum washer spacer	1	4mm thick	Used for mounting the heatsink and 7805 regulator
SCREW1	M3 bolt	2	M3 x 12mm	M3 bolt for fixing the heatsink and 7805 regulator
WASH1	M3 washer	2		Steel washer M3 to assemble heatsink and 7805 regulator
SDCARD	SD-Card for the DFPlayer Mini	1		>=2GB is recommended
BTNCAP	Tactile Push Button Switch Cap 10mm	2		Tactile Push Button Switch Cap 10mm
SCREW2	3 x 12mm Countersunk head screw	4		Used to fix the lid and secure the pcb in the case
SCREW3	M3 Bolt	2	M3 x 12mm	M3 bolt for fixing the classic Dutch PTT phone wall box
WASH2	M3 Washer	2	M3	Steel washer M3 to firmly assemble the wall box
M3NUT	M3 Nut	2	M3	DIN934 M3 Hex Nut Metric Thread
WIRE	Small length of colored (red/blue) electric wire	2	100mm	General purpose 24 AWG electric wire for connecting a/b of phone
PHONE CONN	Landline wall box for connecting a Dutch PTT phone	1		Dutch: Q-Link telefoon contactdoos opbouw (Art.Nr. 00.133.44)
	OR: Wired 6P4C RJ11/12 chassis connector female	1		Use only green and red wires to connect to 'A' and 'B' inputs
POWER	AC/DC 9V 2A Switching Power Supply Adapter	1		AC/DC 9V 2A Switching Power Supply Adapter

IDK "Impuls Druk kiezen": Pulses during the dialing

TDK "Toon Druktoets Keuze": Tones during the dialing